Day 12 — Galaxy Workflow: QC → Trim → Assemble → Evaluate (and Save for Git)

BIOL 4810

2026-03-05

Day 12 — Galaxy Workflow (and saving outputs for Git)

Goal: Run quality control, trimming, assembly, and assembly evaluation using Galaxy, then download/rename key outputs and document a reproducible workflow you can commit to Git.
Workflow: FASTQC → FASTP → FASTQC → SPAdes → QUAST

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0) Account, upload & folder setup (do this first)

A) Create your Galaxy account

1. Go to **https://usegalaxy.org (https://usegalaxy.org)**
2. Create an account (or log in if you already have one) and confirm your email.
3. In Galaxy, create a clear history name for today's work (e.g., `BIOL4810_day12_<yourname>`).

0A. Galaxy username/email used (for your notes):

miguelperez1986

B) Upload reads to Galaxy

1. Click **Upload Data** → choose your FASTQ files (paired-end: R1 and R2).
2. Set a sensible dataset name (include your sampleID). Example: `Sample1_raw_R1.fastq.gz` .
3. Wait for uploads to finish before moving on.

0B. Raw dataset name(s) you uploaded:

WT_R1.fastq.gz; WT_R2.fastq.gz

C) Create your local folder (for Git later)

Create a folder inside your BIOL 4810 course folder (from Unix/Git day), for example:

`module2_wgs/day12_galaxy_qc_assembly/`

0C. Local folder path you will use:

biol4810/module02/AssemblyforAssessment/

File naming convention (required)

When you download files from Galaxy, rename them using:

`<sampleID>_<step>_<tool>_<date>.<ext>`

Example: `Ecoli1_raw_fastqc_2026-03-02.html`

Output	What to download from Galaxy	Rename example
FASTQC raw	FASTQC HTML report (raw reads)	`Sample1_raw_fastqc.html`
FASTP report	fastp HTML/JSON report (optional to keep)	`Sample1_fastp_report.html`
FASTQC trimmed	FASTQC HTML report (trimmed reads)	`Sample1_trimmed_fastqc.html`
Assembly	SPAdes contigs FASTA	`Sample1_spades_contigs.fasta`
QUAST	QUAST **PDF** report (required)	`Sample1_quast_report.pdf`
Workflow notes	your own Markdown file	`workflow_day12.md`

0D. What sampleID will you use for naming (one short ID)?

Sample1

Step 1 — Initial quality check (FASTQC on raw reads)

In Galaxy: 1. Run **FASTQC** on your raw reads.
2. Inspect per-base quality plots and adapter content.

Questions (FASTQC — raw)

1A. Read length (e.g., 150 bp):

151

1B. Does quality drop toward the end of reads? What pattern do you see?

The quality stays pretty consistent. A little wonky towards the beginning but nothing too significantly detrimental to quality.

1C. Are adapter sequences detected or suspected?

None, the line is flat at 0.

Upload screenshot: FASTQC quality plot (raw)

Choose File Screensho….33.16 AM

Step 2 — Trim and filter (fastp)

In Galaxy: 1. Run **fastp** on your raw reads (default settings are fine unless instructor says otherwise).
2. Save the **trimmed FASTQ** output and the fastp report (HTML/JSON) for your records.

Questions (fastp)

2A. What did fastp remove/trim? (adapters, low-quality tails, short reads, etc.)

It removed/trimmed passed filters (3.088820 M), those with low quality (13.190000 K), those with too many N (462), those that were too short (22.49400 K), and those with adapter dimer (60).

2B. Did read length / counts change after trimming? If so, how?

It went from having 151 bp read to 148 bp after filtering

(You do not need to upload the fastp HTML; keep it for your workflow folder.)

Step 3 — FASTQC again (after trimming)

In Galaxy: 1. Run **FASTQC** on the trimmed reads.
2. Inspect the report.

Questions (FASTQC — trimmed)

3A. Did overall quality improve compared to raw?

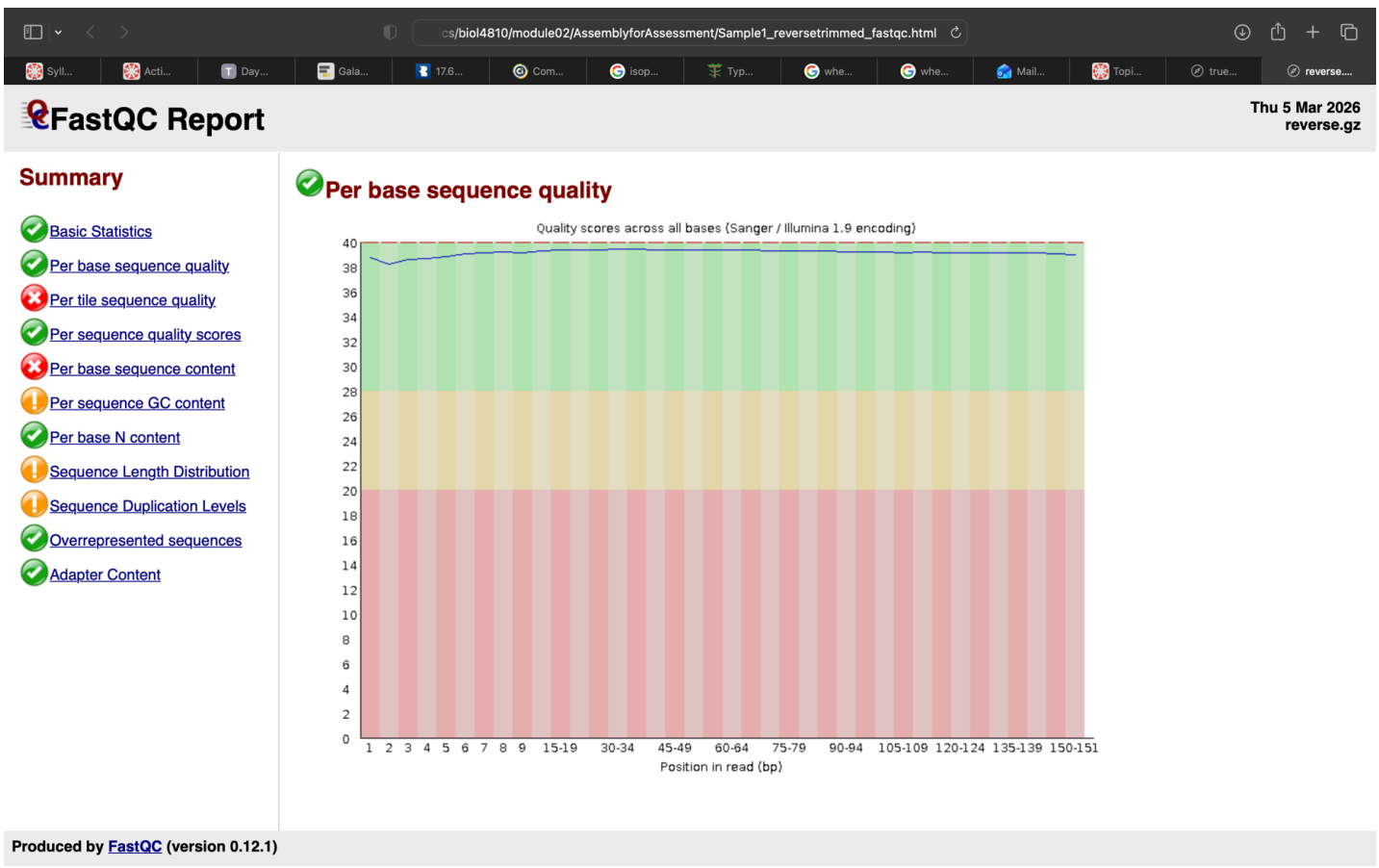
Overall quality did improve as represented by the overrepresented sequence being accounted for and trimmed.

3B. Are adapters still present?

Adaptor content is still at 0.

Upload screenshot: FASTQC quality plot (trimmed)

Choose File Screensho….33.31 AM



Step 4 — Assembly (SPAdes on trimmed reads)

In Galaxy: 1. Run **SPAdes** using your trimmed reads (paired-end if available).
2. Download the contigs FASTA file.

Questions (SPAdes — note: summary metrics come from QUAST)

4A. What is the filename you downloaded from SPAdes?

Sample1_Galaxy52-[SPAdes on dataset 40 and 41 and collection 37_ Contigs].fasta

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Step 5 — Evaluation (QUAST)

In Galaxy: 1. Run **QUAST** on your SPAdes contigs.
2. Download the **QUAST PDF report** (required for submission).

Record (QUAST)

5A. N50:

193475

5B. L50:

8

5C. Number of contigs:

79

5D. Total assembly length (bp):

4797060

Upload QUAST PDF report (required)

Choose File Galaxy73-[…eport].pdf

Selected file: Galaxy73-[Quast on dataset 64_ PDF report].pdf

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Interpretation (short answers)

6A. Did trimming likely improve the assembly? Why or why not?

While trimming might have improved the quality of the read, it may have removed enough of the sequence to shift and change the reading frame.

6B. Is this assembly fragmented? Which metric(s) support your answer?

With the removal of 79 contains, its likely to be fragmented.

Reproducibility checklist (download + rename + document)

A) Download and rename these files into your local folder

Check each item after you have downloaded **and renamed** it:

- [x] **FASTQC (raw) report downloaded + renamed**
- [x] **fastp report downloaded + renamed (recommended)**
- [x] **FASTQC (trimmed) report downloaded + renamed**
- [x] **SPAdes contigs FASTA downloaded + renamed**
- [x] **QUAST PDF report downloaded + renamed**
- [x] **`workflow_day12.md` created in your folder**

B) Record tool versions (from Galaxy Job Info)

Click a dataset → view details/job info and record version numbers.

FASTQC version:

Galaxy Version 0.74+galaxy1

fastp version:

Galaxy Version 1.1.0+galaxy0

SPAdes version:

Galaxy Version 4.2.0+galaxy0

QUAST version:

Galaxy Version 5.3.0+galaxy1

C) Write your workflow notes (copy/paste into `workflow_day12.md`)

Use your sampleID and versions.

```
# Day 12 — Galaxy QC → Trim → Assembly → QUAST

## Sample
- sampleID: YOUR_SAMPLEID
- raw data: (where it came from / dataset name)

## Tools (Galaxy / usegalaxy.org)
- FastQC: VERSION
- fastp: VERSION
- SPAdes: VERSION
- QUAST: VERSION

## Steps
1. FastQC on raw reads → saved report as: YOUR_SAMPLEID_raw_fastqc.html
2. fastp trimming (default settings) → saved report as:
YOUR_SAMPLEID_fastp_report.html
3. FastQC on trimmed reads → saved report as: YOUR_SAMPLEID_trimmed_fastqc.html
4. SPAdes assembly on trimmed reads → saved contigs as:
YOUR_SAMPLEID_spades_contigs.fasta
5. QUAST evaluation on contigs → saved report as: YOUR_SAMPLEID_quast_report.pdf

## Notes / interpretation
- (1–3 bullets on what changed after trimming)
- (1–3 bullets on what QUAST suggests about contiguity)
```

Export (PDF) & Submission

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.
3. Commit your downloaded/renamed files + `workflow_day12.md` to your Git repo.

Submission preview

Setup

Galaxy account: miguelperez1986
Raw datasets: WT_R1.fastq.gz; WT_R2.fastq.gz
Local folder: biol4810/module02/AssemblyforAssessment/
sampleID: Sample1

FASTQC (raw):

```
Read length: 151
Quality pattern: The quality stays pretty consistent. A little wonky towards the
beginning but nothing too significantly detrimental to quality.
Adapters: None, the line is flat at 0.
```

fastp:

```
Removed/trimmed: It removed/trimmed passed filters (3.088820 M), those with low
quality (13.190000 K), those with too many N (462), those that were too short
(22.49400 K), and those with adapter dimer (60).
Read counts change: It went from having 151 bp read to 148 bp after filtering
```

FASTQC (trimmed):

```
Quality improvement: Overall quality did improve as represented by the
overrepresented sequence being accounted for and trimmed.
Adapters after trimming: Adaptor content is still at 0.
```

SPAdes:

```
Downloaded contigs file: Sample1_Galaxy52-[SPAdes on dataset 40 and 41 and collection
37_ Contigs].fasta
```

QUAST:

```
N50: 193475
L50: 8
# contigs: 79
Total length: 4797060
```

Interpretation:

```
Trimming effect: While trimming might have improved the quality of the read, it may
have removed enough of the sequence to shift and change the reading frame.

Fragmentation: With the removal of 79 contains, its likely to be fragmented.
```

Versions (from Galaxy Job Info):

```
FastQC: Galaxy Version 0.74+galaxy1
fastp: Galaxy Version 1.1.0+galaxy0
SPAdes: Galaxy Version 4.2.0+galaxy0
QUAST: Galaxy Version 5.3.0+galaxy1
```

Reproducibility checklist:

```
FASTQC raw downloaded+renamed: Yes
fastp report downloaded+renamed: Yes
FASTQC trimmed downloaded+renamed: Yes
SPAdes contigs downloaded+renamed: Yes
QUAST PDF downloaded+renamed: Yes
workflow_day12.md created: Yes
```